

Classical Mechanics 1, Autumn 2025 CMI

Assignment 2

Due by 6pm Friday 22 Aug, 2025

Levi-Civita symbol, spherical polar coordinates

1. **⟨4 + 3⟩** (a) Starting from $\hat{x} \times \hat{y} = \hat{z}$ and its cyclic permutations, express the Cartesian components of the cross product $\mathbf{a} \times \mathbf{b}$ in terms of those of $\mathbf{a} = a_x\hat{x} + a_y\hat{y} + a_z\hat{z}$ and $\mathbf{b} = b_x\hat{x} + b_y\hat{y} + b_z\hat{z}$. (b) Use this to obtain an expression for the scalar triple product $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ in terms of the components of $\mathbf{a}, \mathbf{b}, \mathbf{c}$. Mention a geometric interpretation of the scalar triple product.
2. **⟨3⟩** Define the Levi-Civita symbol (a real number) ϵ_{ijk} for integers $1 \leq i, j, k \leq 3$ (not necessarily distinct) by the conditions (i) $\epsilon_{123} = 1$ and (ii) ϵ_{ijk} reverses sign under exchange of any pair of indices [e.g. $\epsilon_{ijk} = -\epsilon_{jik}$]. Find ϵ_{213} , ϵ_{312} , ϵ_{321} , ϵ_{112} , ϵ_{313} and ϵ_{222} .
3. **⟨4⟩** Verify that $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \sum_{1 \leq i, j, k \leq 3} \epsilon_{ijk} a_i b_j c_k$ where a_i, b_j, c_k are the Cartesian components of three vectors in $\mathbb{R}^3$. Here $a_1 = a_x, a_2 = a_y, a_3 = a_z$, etc.
4. **⟨4 + 4⟩** Recall the spherical polar coordinates in 3d $z = r \cos \theta, x = r \sin \theta \cos \phi$ and $y = r \sin \theta \sin \phi$. Let $\hat{r}, \hat{\theta}, \hat{\phi}$ be defined as

$$\begin{aligned}\hat{r} &= \cos \theta \hat{z} + \sin \theta (\cos \phi \hat{x} + \sin \phi \hat{y}), \\ \hat{\theta} &= -\sin \theta \hat{z} + \cos \theta (\cos \phi \hat{x} + \sin \phi \hat{y}) \quad \text{and} \\ \hat{\phi} &= -\sin \phi \hat{x} + \cos \phi \hat{y}.\end{aligned}\tag{1}$$

- (a) With the help of figures (3d and suitable projections), explain why the formula for $\hat{\theta}$ gives a unit vector in the direction of increasing θ holding r, ϕ fixed. (b) Verify that $(\hat{r}, \hat{\theta}, \hat{\phi})$ is a right-handed orthonormal system.